Abstract Algebra

Yifan Li

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Introduction

Abstract algebra studies mathematical objects through the *structures* they carry rather than through their concrete representations. Instead of asking whether an object consists of numbers, matrices, permutations, or polynomials, we ask which operations are defined on it and which algebraic laws those operations satisfy.

This point of view allows many apparently different mathematical objects to be treated in the same language. For example, the integers under addition, invertible matrices under multiplication, and permutations under composition are very different kinds of objects, but each of them forms a group. Similarly, the integers, polynomial rings, and matrix rings all possess two compatible operations and are naturally studied as rings.

The basic philosophy of abstract algebra can therefore be summarized schematically as

$$\text{objects} \longrightarrow \text{operations} \longrightarrow \text{algebraic laws} \longrightarrow \text{structure}.$$

Once the underlying structure has been identified, one can study maps that preserve it, substructures contained in it, quotient structures obtained from it, and decompositions that reveal its internal form.

The three principal structures considered in these notes are groups, rings, and fields. A group captures the algebra of a single invertible operation. A ring introduces a second operation and studies the interaction between addition and multiplication. A field is a particularly well-behaved commutative ring in which every nonzero element is multiplicatively invertible.

These structures are not independent topics. They form a natural progression:

$$\text{groups} \longrightarrow \text{rings} \longrightarrow \text{fields}.$$

The additive structure of a ring is itself an abelian group, while the nonzero elements of a field form an abelian group under multiplication. Many ideas first encountered in group theory therefore reappear later in ring theory and field theory in more refined forms.

Another central theme is the study of structure-preserving maps. Homomorphisms tell us when algebraic information can be transported from one object to another, while isomorphisms identify two objects that are structurally indistinguishable. Quotient constructions and isomorphism theorems then provide systematic ways to simplify an algebraic object without losing the information relevant to its structure.

The group-theoretic part of these notes develops subgroups, cosets, normal subgroups, quotient groups, homomorphisms, direct products, permutation groups, cyclic groups, and finite abelian groups. Particular attention is given to automorphism groups of cyclic groups, which show how the internal arithmetic of the order of a group can control the structure of its symmetries.

The ring-theoretic part studies ideals, quotient rings, polynomial rings, integral domains, irreducibility, and factorization. The hierarchy

$$\text{Euclidean domain} \Longrightarrow \text{principal ideal domain} \Longrightarrow \text{unique factorization domain} \Longrightarrow \text{integral domain}$$

organizes much of the discussion and explains which properties of integer factorization survive in more general algebraic settings.

Finally, field theory studies extensions of fields and the algebraic information carried by polynomial equations. Minimal polynomials, extension degrees, splitting fields, and field automorphisms lead naturally to Galois theory, where the symmetries of the roots of a

polynomial are encoded by a group. In this way, the three main parts of the course eventually reconnect:

$$\text{polynomials} \longrightarrow \text{field extensions} \longrightarrow \text{automorphisms} \longrightarrow \text{groups}.$$

The purpose of these notes is therefore not merely to collect definitions and theorems, but to emphasize the structural connections between them. The recurring questions are: what structure is present, which maps preserve it, which substructures are important, how can the object be decomposed, and what invariants remain unchanged under isomorphism?

Chapter 1

Algebraic Structures

1.1 Sets and Operations

Abstract algebra begins with sets equipped with operations. The elements of the set may be numbers, matrices, polynomials, permutations, or other mathematical objects. What matters is not their particular representation, but how the prescribed operations behave.

Definition 1.1.1. Let A be a set. A *binary operation* on A is a map

$$* : A \times A \longrightarrow A.$$

Thus, for every $a, b \in A$, the element $a * b$ must also belong to A.

The requirement that $a * b \in A$ is usually called *closure*. It is essential: an operation on a set must remain inside that set.

Example 1.1.2. Addition and multiplication are binary operations on $\mathbb{Z}$, since for $a, b \in \mathbb{Z}$, both $a + b$ and ab are integers.
Division is not a binary operation on $\mathbb{Z}$, since, for example, $1/2 \notin \mathbb{Z}$.

The same idea applies to more complicated objects. Let F be a field. Then $M_n(F)$, the set of all $n \times n$ matrices with entries in F, is closed under matrix addition and multiplication. Hence both operations are binary operations on $M_n(F)$.
Similarly, the polynomial set

$$F[x] = \{a_0 + a_1 x + \cdots + a_n x^n : n \geq 0,\ a_i \in F\}$$

is closed under polynomial addition and multiplication.

1.2 What Is an Algebraic Structure?

A set by itself contains very little algebraic information. Algebra appears when the set is equipped with one or more operations satisfying specified laws.

Definition 1.2.1. An *algebraic structure* consists of a set together with one or more operations on that set and a collection of axioms describing those operations.

Thus the general pattern is

$$\text{set} + \text{operations} + \text{axioms} = \text{algebraic structure}.$$

For example, the same underlying set may carry different algebraic structures depending on the operation under consideration. The integers $\mathbb{Z}$ behave very differently under addition and multiplication:

$$(\mathbb{Z}, +) \qquad \text{and} \qquad (\mathbb{Z}, \cdot).$$

Therefore, when discussing an algebraic object, it is important to specify not only the underlying set but also its operations.
Three structures will be central throughout these notes:

$$\boxed{\text{group}} \qquad \boxed{\text{ring}} \qquad \boxed{\text{field}}.$$

We introduce them briefly in this chapter. Their deeper structural properties will be developed in the following chapters.

1.3 Groups

A group is the fundamental algebraic structure associated with a single operation.

Definition 1.3.1. A set G equipped with a binary operation $*$ is called a *group* if the following conditions hold:

1. **Associativity.** For all $a, b, c \in G$,
$$a * (b * c) = (a * b) * c.$$
2. **Identity.** There exists $e \in G$ such that
$$a * e = e * a = a$$
for every $a \in G$.
3. **Inverse.** For every $a \in G$, there exists $b \in G$ such that
$$a * b = b * a = e.$$
The element b is called the *inverse* of a.

Notice that closure is already included in the statement that $*$ is a binary operation on G.

Definition 1.3.2. A group G is called *abelian* if
$$a * b = b * a$$
for all $a, b \in G$.

Example 1.3.3. The pair $(\mathbb{Z}, +)$ is an abelian group. Its identity is 0, and the inverse of a is $-a$.
However, $(\mathbb{Z}, \cdot)$ is not a group because most integers do not have multiplicative inverses in $\mathbb{Z}$.

Example 1.3.4. The additive group $(M_n(\mathbb{R}), +)$ is abelian. In contrast, $M_n(\mathbb{R})$ is not a group under multiplication because not every matrix is invertible.
If
$$GL_n(F) = \{A \in M_n(F) : \det A \neq 0\},$$
then $(GL_n(F), \cdot)$ is a group, called the *general linear group*. For $n \geq 2$, it is generally non-abelian.

Groups will be studied systematically in Chapter 2.

1.4 Rings

A ring carries two operations, usually interpreted as addition and multiplication.

Definition 1.4.1. A set R equipped with operations $+$ and $\cdot$ is called a *ring* if:

1. $(R, +)$ is an abelian group;
2. multiplication is associative:
$$a(bc) = (ab)c$$
for all $a, b, c \in R$;

3. multiplication distributes over addition:

$$a(b+c) = ab + ac, \qquad (a+b)c = ac + bc$$

for all $a, b, c \in R$.

In particular, the additive structure of every ring is already a group. This is the first important connection between the algebraic structures considered in these notes.

Definition 1.4.2. A ring R is called *commutative* if $ab = ba$ for all $a, b \in R$.
If there exists $1 \in R$ such that

$$1a = a1 = a$$

for every $a \in R$, then 1 is called the *multiplicative identity* of R.

Example 1.4.3. The integers $(\mathbb{Z}, +, \cdot)$ form a commutative ring with identity. They do not form a field because a nonzero integer such as 2 does not have a multiplicative inverse in $\mathbb{Z}$.

Example 1.4.4. The polynomial set $F[x]$ is a commutative ring under the usual polynomial addition and multiplication.
The matrix set $M_n(F)$ is also a ring. When $n \geq 2$, however, matrix multiplication is not commutative, so $M_n(F)$ is generally a noncommutative ring.

The structure of rings, especially their ideals, quotient rings, and factorization properties, will be studied in Chapter 3.

1.5 Fields

A field is a ring in which division by every nonzero element is possible.

Definition 1.5.1. A set F equipped with addition and multiplication is called a *field* if:

1. $(F, +)$ is an abelian group with identity 0;
2. $(F \setminus \{0\}, \cdot)$ is an abelian group with identity 1;
3. multiplication distributes over addition:

$$a(b+c) = ab + ac, \qquad (a+b)c = ac + bc$$

for all $a, b, c \in F$.

Thus every nonzero $a \in F$ has a multiplicative inverse $a^{-1} \in F$. Consequently, for $b \neq 0$, division can be defined by $a/b = ab^{-1}$.

Example 1.5.2. The familiar number systems

$$, \qquad \mathbb{R}, \qquad \mathbb{C}$$

are fields under their usual addition and multiplication.
The integers $\mathbb{Z}$ are not a field because nonzero integers do not necessarily have multiplicative inverses in $\mathbb{Z}$.

Finite fields provide another important class of examples. If p is prime, arithmetic modulo p gives the field

$$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}.$$

For instance, in $\mathbb{F}_5$, the element 2 has multiplicative inverse 3, since $2 \cdot 3 \equiv 1 \pmod 5$.
Fields and their extensions will be studied in Chapter 4.

1.6 Groups, Rings, and Fields

The three structures introduced above are closely related rather than three isolated collections of definitions.

A ring R contains an abelian group $(R, +)$. A field F is, in particular, a commutative ring, while its nonzero elements form another abelian group $(F \setminus \{0\}, \cdot)$. Schematically,

$$\text{field} \Longrightarrow \text{commutative ring} \Longrightarrow \text{ring},$$

while group structures appear inside both rings and fields.

The converse implications do not hold. For example,

$$\mathbb{Z}$$

is a commutative ring but not a field, while

$$M_n(F), \qquad n \geq 2,$$

is a ring that is generally not commutative.

This hierarchy explains the order of the following chapters. Group theory provides the first structural language. Ring theory introduces a second compatible operation, and field theory imposes sufficiently strong conditions on multiplication to make division possible.

A recurring principle throughout abstract algebra is that the concrete form of the elements is secondary. What matters is the structure preserved by the operations. Thus,

$$\text{numbers}, \quad \text{matrices}, \quad \text{polynomials}, \quad \text{permutations}$$

may look completely different, yet the same algebraic ideas can apply to all of them.

The remaining chapters develop this structural point of view in the order

$$\boxed{\text{Group Theory} \longrightarrow \text{Ring Theory} \longrightarrow \text{Field Theory}}.$$

Chapter 2

Group Theory

2.1 Groups and Subgroups

We now study groups systematically. Recall that a group G is a set equipped with an associative binary operation, an identity element, and inverses for all elements.
Unless otherwise specified, the group operation will be written multiplicatively. The identity is denoted by e, and the inverse of $g \in G$ by g^{-1}.

Definition 2.1.1. The *order* of a finite group G is the number of its elements, denoted by $|G|$.

Definition 2.1.2. Let G be a group. A subset $H \subseteq G$ is called a *subgroup* of G, denoted by $H \leq G$, if H is itself a group under the operation inherited from G.

Thus the operation on H must be exactly the same as the operation on G. For example, a subset equipped with a different operation is not regarded as a subgroup.

Lemma 2.1.3 (Subgroup Test)**.** Let H be a nonempty subset of a group G. Then $H \leq G$ if and only if

$$xy^{-1} \in H$$

for all $x, y \in H$.

Proof. If $H \leq G$, then $y^{-1} \in H$, so $xy^{-1} \in H$.
Conversely, suppose $H \neq \varnothing$ and $xy^{-1} \in H$ for all $x, y \in H$. Taking $x = y$ gives $e = xx^{-1} \in H$. Taking $x = e$ gives $y^{-1} \in H$. Hence for $x, y \in H$, since $y^{-1} \in H$,

$$x(y^{-1})^{-1} = xy \in H.$$

Therefore H is closed under products and inverses, and hence $H \leq G$. □

For a finite group, the subgroup test can be simplified.

Proposition 2.1.4. Let G be finite and let $H \subseteq G$ be nonempty. If H is closed under multiplication, then $H \leq G$.

Example 2.1.5. Let F be a field. The *general linear group*

$$GL_n(F) = \{A \in M_n(F) : \det A \neq 0\}$$

is a group under matrix multiplication.
The set

$$SL_n(F) = \{A \in GL_n(F) : \det A = 1\}$$

is a subgroup of $GL_n(F)$, called the *special linear group.*

2.1.1 The Center

Definition 2.1.6. The *center* of a group G is

$$Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}.$$

The center measures the part of a group that commutes with every element.

Proposition 2.1.7. For every group G, $Z(G) \leq G$.

In particular,

$$G \text{ is abelian} \iff Z(G) = G.$$

For the general linear group,

$$Z(GL_n(F)) = \{aI_n : a \in F^\times\},$$

where $F^\times = F \setminus \{0\}$.

2.2 Cosets and Lagrange's Theorem

Let $H \leq G$ and $g \in G$.

Definition 2.2.1. The sets

$$Hg = \{hg : h \in H\}, \qquad gH = \{gh : h \in H\}$$

are called the *right coset* and *left coset* of H determined by g, respectively.

Lemma 2.2.2. Let $H \leq G$. For $g_1, g_2 \in G$, two right cosets are either equal or disjoint:

$$Hg_1 = Hg_2 \qquad \text{or} \qquad Hg_1 \cap Hg_2 = \varnothing.$$

Proof. Suppose $Hg_1 \cap Hg_2 \neq \varnothing$. Then there exist $h_1, h_2 \in H$ such that

$$h_1 g_1 = h_2 g_2.$$

Hence

$$g_1 = h_1^{-1} h_2 g_2,$$

and $h_1^{-1} h_2 \in H$. Therefore

$$Hg_1 = Hh_1^{-1} h_2 g_2 = Hg_2.$$

□

For every $g \in G$, the map $H \to Hg$, $h \mapsto hg$, is a bijection. Hence every coset of H has the same cardinality as H.

Definition 2.2.3. The number of left or right cosets of H in G is called the *index* of H in G, denoted by

$$[G : H].$$

Theorem 2.2.4 (Lagrange's Theorem)**.** Let G be a finite group and $H \leq G$. Then

$$|G| = [G : H] \, |H| \, .$$

In particular, $|H| \mid |G|$.

Proof. Let

$$Hg_1, \ldots, Hg_m$$

be all distinct right cosets of H. Since cosets partition G,

$$G = Hg_1 \sqcup \cdots \sqcup Hg_m.$$

Each coset has $|H|$ elements. Therefore

$$|G| = m \, |H| = [G : H] \, |H| \, .$$

□

2.2.1 Order of an Element

Definition 2.2.5. Let $g \in G$. The *order* of g, denoted by $\text{ord}(g)$, is the smallest positive integer n such that $g^n = e$, if such an integer exists.

The subgroup generated by g is

$$\langle g \rangle = \{g^k : k \in \mathbb{Z}\}.$$

If g has finite order n, then

$$\langle g \rangle = \{e, g, g^2, \ldots, g^{n-1}\},$$

so

$$\text{ord}(g) = |\langle g \rangle|.$$

By Lagrange's theorem, if G is finite, then

$$\text{ord}(g) \mid |G|.$$

Corollary 2.2.6. If G is finite, then

$$g^{|G|} = e$$

for every $g \in G$.

Corollary 2.2.7. Every group of prime order is cyclic.

2.2.2 Fermat and Euler Theorems

Lagrange's theorem immediately gives familiar results from elementary number theory.

Theorem 2.2.8 (Fermat's Little Theorem)**.** Let p be prime and $a \not\equiv 0 \pmod{p}$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. The nonzero residue classes modulo p form a multiplicative group of order $p-1$. Apply Lagrange's theorem. □

For a positive integer n, define

$$U(n) = \{\overline{a} \in \mathbb{Z}/n\mathbb{Z} : \gcd(a, n) = 1\}.$$

Then $U(n)$ is a group under multiplication modulo n, with

$$|U(n)| = \varphi(n),$$

where φ is Euler's totient function.

Theorem 2.2.9 (Euler's Theorem)**.** If $\gcd(a, n) = 1$, then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

2.3 Normal Subgroups and Quotient Groups

Ordinary subgroups do not always allow us to form a group of cosets. The correct condition is normality.

Definition 2.3.1. A subgroup $N \leq G$ is called a *normal subgroup*, denoted by $N \trianglelefteq G$, if

$$g^{-1}ng \in N$$

for every $n \in N$ and $g \in G$.

Theorem 2.3.2. Let $N \leq G$. The following statements are equivalent:

1. $N \trianglelefteq G$;
2. $g^{-1}Ng = N$ for every $g \in G$;
3. $g^{-1}ng \in N$ for every $n \in N$ and $g \in G$;
4. $gN = Ng$ for every $g \in G$.

Example 2.3.3. For every field F,

$$SL_n(F) \trianglelefteq GL_n(F).$$

Indeed, if $A \in SL_n(F)$ and $P \in GL_n(F)$, then

$$\det(P^{-1}AP) = \det(P)^{-1}\det(A)\det(P) = 1.$$

2.3.1 Normalizer and Centralizer

Definition 2.3.4. Let $S \subseteq G$. The *normalizer* of S in G is

$$N_G(S) = \{g \in G : g^{-1}Sg = S\}.$$

The *centralizer* of S in G is

$$C_G(S) = \{g \in G : gs = sg \text{ for every } s \in S\}.$$

In particular,

$$Z(G) = C_G(G).$$

2.3.2 Quotient Groups

Let $N \trianglelefteq G$. Define

$$G/N = \{gN : g \in G\}.$$

Theorem 2.3.5. If $N \trianglelefteq G$, then the operation

$$(g_1N)(g_2N) = (g_1g_2)N$$

is well-defined and makes G/N into a group.

Proof. Suppose

$$g_1N = g_1'N, \qquad g_2N = g_2'N.$$

Then $g_1' = g_1n_1$ and $g_2' = g_2n_2$ for some $n_1, n_2 \in N$. Hence

$$g_1'g_2' = g_1n_1g_2n_2 = g_1g_2(g_2^{-1}n_1g_2)n_2.$$

Since $N \trianglelefteq G$, $g_2^{-1} n_1 g_2 \in N$, so

$$g_1' g_2' N = g_1 g_2 N.$$

Thus the multiplication is well-defined. The identity is N, and

$$(gN)^{-1} = g^{-1} N.$$

The remaining group axioms follow from those of G. □

The group G/N is called the *quotient group* or *factor group* of G by N.

2.4 Homomorphisms and Isomorphism Theorems

Definition 2.4.1. Let G and H be groups. A map

$$\varphi : G \longrightarrow H$$

is a *group homomorphism* if

$$\varphi(g_1 g_2) = \varphi(g_1)\varphi(g_2)$$

for all $g_1, g_2 \in G$.

A bijective homomorphism is called an *isomorphism*. If there exists an isomorphism $G \to H$, we write

$$G \cong H.$$

Two isomorphic groups are regarded as the same from the structural point of view.

Definition 2.4.2. Let $\varphi : G \to H$ be a homomorphism. Its *kernel* and *image* are

$$\ker \varphi = \{g \in G : \varphi(g) = e_H\},$$

and

$$\operatorname{Im} \varphi = \{\varphi(g) : g \in G\}.$$

Proposition 2.4.3. For a homomorphism $\varphi : G \to H$,

$$\ker \varphi \trianglelefteq G, \qquad \operatorname{Im} \varphi \leq H.$$

Moreover,

$$\varphi \text{ is injective} \iff \ker \varphi = \{e\}.$$

2.4.1 First Isomorphism Theorem

Theorem 2.4.4 (First Isomorphism Theorem)**.** Let $\varphi : G \to H$ be a group homomorphism. Then

$$G/\ker \varphi \cong \operatorname{Im} \varphi.$$

Proof. Define

$$\overline{\varphi} : G/\ker \varphi \longrightarrow \operatorname{Im} \varphi$$

by

$$\overline{\varphi}(g \ker \varphi) = \varphi(g).$$

If

$$g_1 \ker \varphi = g_2 \ker \varphi,$$

then $g_2^{-1} g_1 \in \ker \varphi$, so

$$\varphi(g_1) = \varphi(g_2).$$

Thus the map is well-defined. It is a homomorphism and is surjective by definition. If $\overline{\varphi}(g \ker \varphi) = e_H$, then $g \in \ker \varphi$, so $g \ker \varphi = \ker \varphi$. Hence the map is injective. □

2.4.2 Second Isomorphism Theorem

For subgroups $A, B \leq G$, define

$$AB = \{ab : a \in A,\ b \in B\}.$$

If $A \trianglelefteq G$, then $AB \leq G$.

Theorem 2.4.5 (Second Isomorphism Theorem)**.** Let $A, B \leq G$ and suppose $A \trianglelefteq G$. Then

$$A \cap B \trianglelefteq B$$

and

$$AB/A \cong B/(A \cap B).$$

Proof. Define

$$\varphi : B \longrightarrow AB/A, \qquad b \longmapsto bA.$$

This is a surjective homomorphism, and

$$\ker \varphi = A \cap B.$$

The result follows from the first isomorphism theorem. □

2.4.3 Third Isomorphism Theorem

Theorem 2.4.6 (Third Isomorphism Theorem)**.** Let

$$H \trianglelefteq K \trianglelefteq G.$$

Then

$$K/H \trianglelefteq G/H$$

and

$$(G/H)/(K/H) \cong G/K.$$

The natural homomorphism is

$$G/H \longrightarrow G/K, \qquad gH \longmapsto gK,$$

whose kernel is K/H.

2.5 Cyclic Groups

Definition 2.5.1. A group G is called *cyclic* if there exists $g \in G$ such that

$$G = \langle g \rangle.$$

The element g is called a *generator* of G.

The infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$, while every finite cyclic group of order n is isomorphic to $\mathbb{Z}/n\mathbb{Z}$.

Proposition 2.5.2. Every subgroup of a cyclic group is cyclic.

Proposition 2.5.3. Let $G = \langle g \rangle$ have order n. Then

$$\operatorname{ord}(g^k) = \frac{n}{\gcd(n, k)}.$$

Consequently, g^k generates G if and only if

$$\gcd(k, n) = 1.$$

Therefore a cyclic group of order n has exactly $\varphi(n)$ generators.

2.5.1 Dihedral Groups

The symmetries of a regular polygon provide a fundamental non-abelian example.

Definition 2.5.4. The *dihedral group* D_{2n} is generated by elements a, b satisfying

$$a^n = e, \qquad b^2 = e, \qquad bab = a^{-1}.$$

Equivalently,

$$D_{2n} = \langle a, b : a^n = b^2 = e,\ bab = a^{-1} \rangle.$$

Its elements are

$$e, a, \ldots, a^{n-1}, b, ab, \ldots, a^{n-1}b,$$

so

$$|D_{2n}| = 2n.$$

The elements $a^k b$ have order 2, corresponding geometrically to reflections.

2.6 Direct Products

Definition 2.6.1. Let G and H be groups. Their *direct product* is

$$G \times H = \{(g, h) : g \in G,\ h \in H\}$$

with multiplication

$$(g_1, h_1)(g_2, h_2) = (g_1 g_2, h_1 h_2).$$

Proposition 2.6.2. If $g \in G$ and $h \in H$ have finite orders, then

$$\operatorname{ord}(g, h) = \operatorname{lcm}\Big(\operatorname{ord}(g), \operatorname{ord}(h)\Big).$$

An important special case is

$$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$$

whenever $\gcd(m, n) = 1$.

2.6.1 Internal Direct Products

Sometimes a group is already built from two of its own subgroups.

Theorem 2.6.3. Let $H, K \trianglelefteq G$ and suppose

$$G = HK.$$

Then the following are equivalent:

1. the map
$$\varphi : H \times K \longrightarrow G, \qquad (h, k) \longmapsto hk$$
is an isomorphism;
2. $$H \cap K = \{e\}.$$

Proof. Suppose first that φ is an isomorphism. If $x \in H \cap K$, then

$$\varphi(x, e) = x = \varphi(e, x).$$

Injectivity gives $x = e$.
Conversely, suppose $H \cap K = \{e\}$. Since both H and K are normal, for $h \in H$ and $k \in K$,

$$hkh^{-1}k^{-1} \in H \cap K.$$

Hence $hk = kh$. Therefore

$$\varphi(h_1, k_1)\varphi(h_2, k_2) = h_1k_1h_2k_2 = h_1h_2k_1k_2,$$

so φ is a homomorphism. It is surjective because $G = HK$.
Finally, if $hk = e$, then $h = k^{-1} \in H \cap K$, so $h = k = e$. Hence the kernel is trivial, and φ is an isomorphism. □

In this case we write

$$G = H \times K$$

and say that G is the *internal direct product* of H and K.

2.7 Permutation Groups

Definition 2.7.1. Let

$$\Omega = \{1, 2, \ldots, n\}.$$

A bijection from Ω to itself is called a *permutation*. The set of all permutations of Ω, with composition as multiplication, is the *symmetric group*

$$S_n = \mathrm{Sym}(\Omega).$$

Since there are $n!$ bijections,

$$|S_n| = n!.$$

Every subgroup of S_n is called a *permutation group*.

2.7.1 Cycle Decomposition

A permutation may be represented by cycles. For example,

$$(1\,6)(2\,3\,4)$$

interchanges 1 and 6, cyclically permutes $2, 3, 4$, and fixes all other points.

Theorem 2.7.2. Every permutation can be written as a product of disjoint cycles.

Disjoint cycles commute.
A cycle

$$(i_1\, i_2\, \ldots\, i_m)$$

can be written as a product of transpositions:

$$(i_1\, i_2\, \ldots\, i_m) = (i_1\, i_2)(i_1\, i_3)\cdots(i_1\, i_m),$$

up to the convention used for composition.
Hence every permutation is a product of transpositions.

2.7.2 Parity and Alternating Groups

Although the decomposition into transpositions is not unique, the parity of the number of transpositions is unique.

Definition 2.7.3. A permutation is called *even* if it can be expressed as a product of an even number of transpositions, and *odd* otherwise.

Definition 2.7.4. The set of all even permutations in S_n is called the *alternating group*, denoted by A_n.

We have

$$A_n \trianglelefteq S_n, \qquad [S_n : A_n] = 2,$$

and therefore

$$|A_n| = \frac{n!}{2}.$$

Every element of A_n can be written as a product of 3-cycles.

2.7.3 Cayley's Theorem

Permutation groups are universal among groups.

Theorem 2.7.5 (Cayley's Theorem)**.** Every group G is isomorphic to a subgroup of $\mathrm{Sym}(G)$. In particular, every finite group of order n is isomorphic to a subgroup of S_n.

Proof. For each $g \in G$, define

$$\lambda_g : G \longrightarrow G, \qquad x \longmapsto gx.$$

Each λ_g is a permutation of G. Define

$$\Phi : G \longrightarrow \mathrm{Sym}(G), \qquad g \longmapsto \lambda_g.$$

Since

$$\lambda_{gh} = \lambda_g \lambda_h,$$

the map Φ is a homomorphism. If λ_g is the identity, then

$$g = \lambda_g(e) = e.$$

Thus $\ker \Phi = \{e\}$, so Φ is injective. □

2.8 Group Actions

Group actions provide a systematic way to study a group through its symmetries of another object.

Definition 2.8.1. Let G be a group and X a set. A *right action* of G on X assigns to every $g \in G$ a bijection $x \mapsto xg$ such that

$$xe = x$$

and

$$(xg)h = x(gh)$$

for all $x \in X$ and $g, h \in G$.

Equivalently, a group action can be described by a homomorphism

$$G \longrightarrow \mathrm{Sym}(X).$$

Important examples include:

- S_n and A_n acting naturally on $\{1, \ldots, n\}$;
- $GL(V)$ acting on a vector space V;
- a group acting on itself by multiplication;
- a group acting on itself by conjugation.

2.8.1 Orbits and Stabilizers

Definition 2.8.2. Let G act on X. The *orbit* of $x \in X$ is

$$xG = \{xg : g \in G\}.$$

The orbits partition X.

Definition 2.8.3. The *stabilizer* of $x \in X$ is

$$G_x = \{g \in G : xg = x\}.$$

Theorem 2.8.4 (Orbit–Stabilizer Theorem)**.** Let a finite group G act on X. Then

$$|G| = |xG| \, |G_x| \, .$$

Equivalently,

$$|xG| = [G : G_x].$$

2.8.2 Conjugation and the Class Equation

Let G act on itself by conjugation:

$$x^g = g^{-1}xg.$$

The orbit of x is its *conjugacy class*

$$\mathrm{Cl}_G(x) = \{g^{-1}xg : g \in G\}.$$

Its stabilizer is the centralizer $C_G(x)$. Hence

$$|\mathrm{Cl}_G(x)| = [G : C_G(x)].$$

The elements with singleton conjugacy classes are precisely the elements of $Z(G)$. Therefore a finite group satisfies the *class equation*

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)],$$

where the x_i represent the noncentral conjugacy classes.

Corollary 2.8.5. Every nontrivial finite p-group has nontrivial center.

2.9 Sylow Theory

Let

$$|G| = p^r m, \qquad \gcd(p, m) = 1,$$

where p is prime.

Definition 2.9.1. A subgroup $P \leq G$ of order p^r is called a *Sylow p-subgroup* of G. The set of all Sylow p-subgroups is denoted by

$$\mathrm{Syl}_p(G).$$

2.9.1 First Sylow Theorem

Theorem 2.9.2 (First Sylow Theorem)**.** If $p^r \mid |G|$, then G contains a subgroup of order p^r. In particular, G contains a Sylow p-subgroup.

Corollary 2.9.3 (Cauchy's Theorem)**.** If $p \mid |G|$, then G contains an element of order p.

2.9.2 Second Sylow Theorem

Theorem 2.9.4 (Second Sylow Theorem)**.** Let $P \in \mathrm{Syl}_p(G)$, and let $H \leq G$ be a p-subgroup. Then H is conjugate to a subgroup of P.
In particular, all Sylow p-subgroups are conjugate.

Consequently,

$$P \trianglelefteq G \quad \Longleftrightarrow \quad P \text{ is the unique Sylow } p\text{-subgroup.}$$

2.9.3 Third Sylow Theorem

Let

$$n_p = \left|\mathrm{Syl}_p(G)\right|.$$

Theorem 2.9.5 (Third Sylow Theorem)**.** If

$$|G| = p^r m, \qquad \gcd(p, m) = 1,$$

then

$$n_p \mid m$$

and

$$n_p \equiv 1 \pmod{p}.$$

These numerical restrictions often determine the possible structure of a finite group.

Example 2.9.6. Suppose $|G| = pq$, where $p > q$ are primes. Then

$$n_p \mid q, \qquad n_p \equiv 1 \pmod{p}.$$

Since $q < p$, the only possibility is $n_p = 1$. Therefore the Sylow p-subgroup is normal in G.

2.10 Finite Abelian Groups

The structure of finite abelian groups can be completely classified.

Theorem 2.10.1 (Fundamental Theorem of Finite Abelian Groups)**.** Every finite abelian group is a direct product of cyclic groups of prime-power order.

More precisely, if

$$|G| = p_1^{a_1} \cdots p_r^{a_r},$$

then

$$G \cong G_{p_1} \times \cdots \times G_{p_r},$$

where G_{p_i} consists of the elements whose orders are powers of p_i.
Each G_{p_i} can in turn be decomposed as

$$G_{p_i} \cong \mathbb{Z}_{p_i^{b_1}} \times \mathbb{Z}_{p_i^{b_2}} \times \cdots \times \mathbb{Z}_{p_i^{b_s}}.$$

Thus the classification problem reduces to finite abelian p-groups.

Example 2.10.2. Every group of order p^2 is abelian and is isomorphic to exactly one of

$$\mathbb{Z}_{p^2}, \qquad \mathbb{Z}_p \times \mathbb{Z}_p.$$

2.11 Automorphism Groups of Cyclic Groups

The automorphism group of a cyclic group provides a useful meeting point between group theory and elementary number theory.

Definition 2.11.1. An *automorphism* of a group G is an isomorphism

$$\sigma : G \longrightarrow G.$$

The set of all automorphisms of G, under composition, is denoted by

$$\mathrm{Aut}(G).$$

The key observation for cyclic groups is simple: an automorphism is completely determined by the image of a generator, and a generator must be mapped to another generator.

2.11.1 The Infinite Cyclic Group

Consider $G = \mathbb{Z}$. Its only generators are 1 and -1. Hence an automorphism must satisfy

$$\sigma(1) = 1 \qquad \text{or} \qquad \sigma(1) = -1.$$

Therefore

$$\mathrm{Aut}(\mathbb{Z}) \cong \mathbb{Z}_2.$$

2.11.2 Finite Cyclic Groups

Let

$$G = \langle a \rangle$$

be cyclic of order n. Every automorphism has the form

$$\sigma_k(a) = a^k,$$

where $\gcd(k, n) = 1$. Moreover,

$$\sigma_k \sigma_\ell = \sigma_{k\ell}.$$

Hence

$$\mathrm{Aut}(\mathbb{Z}_n) \cong U(n) = (\mathbb{Z}/n\mathbb{Z})^\times.$$

In particular,

$$|\mathrm{Aut}(\mathbb{Z}_n)| = \varphi(n).$$

The remaining question is therefore not the order but the *structure* of $U(n)$.

2.11.3 The Case of Prime Order

Let p be prime. Then

$$|\mathrm{Aut}(\mathbb{Z}_p)| = \varphi(p) = p - 1.$$

Theorem 2.11.2. Every prime p has a primitive root modulo p.

Choose a primitive root k modulo p. Then

$$\sigma_k(a) = a^k$$

has order $p - 1$. Hence

$$\mathrm{Aut}(\mathbb{Z}_p) \cong \mathbb{Z}_{p-1}.$$

This suggests the general principle that the order of $\mathrm{Aut}(\mathbb{Z}_n)$ is controlled by $\varphi(n)$, while its group structure is closely related to the existence of primitive roots modulo n.

2.11.4 Odd Prime Powers

Let p be an odd prime.

Theorem 2.11.3. For every positive integer r, the integer p^r has a primitive root.

Therefore

$$U(p^r)$$

contains an element of order

$$\varphi(p^r) = p^{r-1}(p - 1).$$

Since this is the order of the entire group,

$$U(p^r) \cong \mathbb{Z}_{p^{r-1}(p-1)}.$$

Consequently,

$$\boxed{\mathrm{Aut}(\mathbb{Z}_{p^r}) \cong \mathbb{Z}_{p^{r-1}(p-1)}}$$

for every odd prime p.

2.11.5 The Exceptional Case $p = 2$

For small powers of 2,

$$\mathrm{Aut}(\mathbb{Z}_2) = \{e\},$$

and

$$\mathrm{Aut}(\mathbb{Z}_4) \cong \mathbb{Z}_2.$$

The first genuinely different case occurs for $\mathbb{Z}_8$. Its automorphisms are

$$\sigma_1, \sigma_3, \sigma_5, \sigma_7,$$

where

$$\sigma_k(a) = a^k.$$

We have

$$\sigma_3^2 = \sigma_5^2 = \sigma_7^2 = e.$$

Thus there is no element of order 4, and hence

$$\mathrm{Aut}(\mathbb{Z}_8) \ncong \mathbb{Z}_4.$$

Instead,

$$\mathrm{Aut}(\mathbb{Z}_8) \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The phenomenon persists for higher powers of 2.

Theorem 2.11.4. For $r \geq 3$, 2^r has no primitive root.

Thus $U(2^r)$ is not cyclic for $r \geq 3$.
Nevertheless, the element 5 provides a large cyclic subgroup.

Theorem 2.11.5. For $r \geq 3$,

$$\mathrm{ord}_{2^r}(5) = 2^{r-2}.$$

Hence

$$\langle \sigma_5 \rangle$$

has order 2^{r-2}.
Now consider

$$\sigma_{-1}(a) = a^{-1}.$$

It has order 2.

Proposition 2.11.6. For $r \geq 3$,

$$\sigma_{-1} \notin \langle \sigma_5 \rangle.$$

Proof. If $\sigma_{-1} \in \langle \sigma_5 \rangle$, then because σ_{-1} has order 2, we would have

$$5^{2^{r-3}} \equiv -1 \pmod{2^r}.$$

However,

$$5^{2^{r-3}} \equiv 1 + 2^{r-1} \pmod{2^r},$$

which is not congruent to $-1 \pmod{2^r}$. □

Since σ_5 and σ_{-1} commute and their generated subgroups intersect trivially,

$$\mathrm{Aut}(\mathbb{Z}_{2^r}) = \langle \sigma_5 \rangle \times \langle \sigma_{-1} \rangle.$$

Therefore, for $r \geq 3$,

$$\boxed{\mathrm{Aut}(\mathbb{Z}_{2^r}) \cong \mathbb{Z}_{2^{r-2}} \times \mathbb{Z}_2}.$$

2.11.6 The General Finite Cyclic Group

Let

$$n = p_1^{r_1} p_2^{r_2} \cdots p_s^{r_s}$$

be the prime factorization of n.
Since distinct prime powers are coprime,

$$\mathbb{Z}_n \cong \mathbb{Z}_{p_1^{r_1}} \times \mathbb{Z}_{p_2^{r_2}} \times \cdots \times \mathbb{Z}_{p_s^{r_s}}.$$

Theorem 2.11.7. If $\gcd(m, n) = 1$, then

$$\mathrm{Aut}(\mathbb{Z}_m \times \mathbb{Z}_n) \cong \mathrm{Aut}(\mathbb{Z}_m) \times \mathrm{Aut}(\mathbb{Z}_n).$$

One way to see the decomposition is to use the natural projections and inclusions

$$\rho_1 : \mathbb{Z}_m \times \mathbb{Z}_n \longrightarrow \mathbb{Z}_m, \qquad \rho_2 : \mathbb{Z}_m \times \mathbb{Z}_n \longrightarrow \mathbb{Z}_n,$$

and

$$i_1 : \mathbb{Z}_m \longrightarrow \mathbb{Z}_m \times \mathbb{Z}_n, \qquad i_2 : \mathbb{Z}_n \longrightarrow \mathbb{Z}_m \times \mathbb{Z}_n.$$

For $\sigma \in \mathrm{Aut}(\mathbb{Z}_m \times \mathbb{Z}_n)$, define

$$\Phi(\sigma) = (\rho_1 \sigma i_1, \rho_2 \sigma i_2).$$

The coprimality condition forces the two primary components to be preserved, and Φ is an isomorphism.
Consequently,

$$\boxed{\mathrm{Aut}(\mathbb{Z}_n) \cong \prod_{i=1}^{s} \mathrm{Aut}\left(\mathbb{Z}_{p_i^{r_i}}\right)}.$$

Together with the prime-power cases, this completely determines the automorphism group of every finite cyclic group.

2.12 Simple and Solvable Groups

Definition 2.12.1. A nontrivial group G is called *simple* if its only normal subgroups are

$$\{e\} \qquad \text{and} \qquad G.$$

Simple groups play a role analogous to prime numbers: they cannot be decomposed further through nontrivial quotient constructions.

2.12.1 Commutator Subgroups

Definition 2.12.2. For $x, y \in G$, the element

$$[x, y] = x^{-1} y^{-1} x y$$

is called the *commutator* of x and y.

Definition 2.12.3. The subgroup generated by all commutators,

$$G' = \langle [x, y] : x, y \in G \rangle,$$

is called the *commutator subgroup* or *derived subgroup* of G.

Proposition 2.12.4. The commutator subgroup is normal:

$$G' \trianglelefteq G.$$

Moreover,

$$G/G'$$

is abelian.

More generally, if $N \trianglelefteq G$, then

$$G/N \text{ is abelian} \quad \Longleftrightarrow \quad G' \leq N.$$

Thus G' is the smallest normal subgroup whose quotient is abelian.

2.12.2 Derived Series

Define recursively

$$G^{(0)} = G, \qquad G^{(1)} = G', \qquad G^{(n+1)} = \left(G^{(n)}\right)'.$$

This gives the derived series

$$G \trianglerighteq G' \trianglerighteq G'' \trianglerighteq \cdots .$$

Definition 2.12.5. A finite group G is called *solvable* if

$$G^{(n)} = \{e\}$$

for some $n \geq 0$.

Every abelian group is solvable, since $G' = \{e\}$.

Theorem 2.12.6. A finite group G is solvable if and only if there exists a chain

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = \{e\}$$

such that every quotient

$$G_i/G_{i+1}$$

is abelian.

2.12.3 Simplicity of A_n

Theorem 2.12.7. The alternating group A_n is simple for $n \geq 5$.

A central observation in the proof is that every element of A_n is a product of 3-cycles, and all 3-cycles are conjugate inside A_n.
Suppose

$$\{e\} \neq N \trianglelefteq A_n.$$

One first shows that N must contain a 3-cycle. Since N is normal, it then contains every conjugate of this 3-cycle. Therefore N contains all 3-cycles. Since 3-cycles generate A_n,

$$N = A_n.$$

Thus A_n has no nontrivial proper normal subgroup.
In particular, A_5 is a non-abelian simple group. This fact will later become important in Galois theory when solvability of groups is connected with solvability of polynomial equations by radicals.

2.13 Composition Series

Definition 2.13.1. A chain

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = \{e\}$$

is called a *subnormal series* if

$$G_{i+1} \trianglelefteq G_i$$

for every i.

Definition 2.13.2. A subnormal series is called a *composition series* if every factor

$$G_i/G_{i+1}$$

is simple. These quotient groups are called the *composition factors* of G.

Example 2.13.3. Consider

$$S_4 \trianglerighteq A_4 \trianglerighteq V_4 \trianglerighteq \mathbb{Z}_2 \trianglerighteq \{e\}.$$

The corresponding composition factors are isomorphic to

$$\mathbb{Z}_2, \qquad \mathbb{Z}_3, \qquad \mathbb{Z}_2, \qquad \mathbb{Z}_2.$$

Theorem 2.13.4 (Jordan–Hölder Theorem)**.** For a finite group G, the composition factors are uniquely determined by G, up to isomorphism and rearrangement.

Thus a group may admit different composition series, but the simple groups appearing as its composition factors are intrinsic.

2.14 Semidirect Products and Group Extensions

Direct products are not sufficient to describe all groups built from smaller groups. Semidirect products allow one factor to act nontrivially on another.

2.14.1 Short Exact Sequences

A sequence

$$1 \longrightarrow K \xrightarrow{\iota} G \xrightarrow{\pi} H \longrightarrow 1$$

is called a *short exact sequence* if

$$\ker \pi = \operatorname{Im} \iota,$$

with ι injective and π surjective.
In this situation,

$$H \cong G/K,$$

and G is called an *extension* of H by K.

Definition 2.14.1. A map

$$s : H \longrightarrow G$$

is called a *section* if

$$\pi \circ s = \mathrm{id}_H .$$

If s is also a homomorphism, the extension is called a *split extension.*

For a split extension,

$$s(H) \leq G, \qquad K \cap s(H) = \{e\},$$

and

$$G = Ks(H).$$

2.14.2 Semidirect Products

Let K and H be groups and let

$$\varphi : H \longrightarrow \mathrm{Aut}(K)$$

be a homomorphism.

Definition 2.14.2. The *semidirect product* of K by H with respect to φ, denoted by

$$K \rtimes_\varphi H,$$

is the set $K \times H$ with multiplication

$$(k_1, h_1)(k_2, h_2) = (k_1\varphi(h_1)(k_2), h_1h_2)\,.$$

Theorem 2.14.3. For any groups K, H and any homomorphism

$$\varphi : H \longrightarrow \mathrm{Aut}(K),$$

the semidirect product

$$K \rtimes_\varphi H$$

exists. Conversely, every split extension

$$1 \longrightarrow K \longrightarrow G \longrightarrow H \longrightarrow 1$$

is isomorphic to a semidirect product of K and H.

The action in a split extension arises naturally from conjugation:

$$\varphi(h)(k) = hkh^{-1}.$$

If this action is trivial, then

$$K \rtimes_\varphi H \cong K \times H.$$

Thus the direct product is the special case of the semidirect product in which the two factors do not interact.

Example 2.14.4. The dihedral group can be expressed as

$$D_{2n} \cong \mathbb{Z}_n \rtimes \mathbb{Z}_2,$$

where the nontrivial element of $\mathbb{Z}_2$ acts on $\mathbb{Z}_n$ by inversion:

$$a \longmapsto a^{-1}.$$

2.14.3 Holomorphs

Definition 2.14.5. The *holomorph* of a group G is

$$\mathrm{Hol}(G) = G \rtimes \mathrm{Aut}(G),$$

where $\mathrm{Aut}(G)$ acts naturally on G.

Thus the holomorph combines the elements of G with all of its automorphisms into a single group.
For a cyclic group, the description of $\mathrm{Aut}(G)$ obtained earlier therefore gives direct information about $\mathrm{Hol}(G)$. This also illustrates why understanding automorphism groups is essential when classifying semidirect products and group extensions.

Chapter 3

Ring Theory

3.1 Rings and Basic Properties

Recall that a ring R is a set equipped with addition and multiplication such that $(R, +)$ is an abelian group, multiplication is associative, and multiplication distributes over addition.
Throughout this chapter, multiplication will usually be written without a symbol.

Definition 3.1.1. A ring R is called *commutative* if

$$ab = ba$$

for all $a, b \in R$.

Definition 3.1.2. An element $1 \in R$ is called a *multiplicative identity* if

$$1a = a1 = a$$

for every $a \in R$.

Unless otherwise specified, many of the rings considered below will be commutative rings with identity.

3.1.1 Left and Right Inverses

Let $a, b \in R$. If

$$ba = 1,$$

then b is called a *left inverse* of a. If

$$ab = 1,$$

then b is called a *right inverse* of a.
If

$$ab = ba = 1,$$

then b is called the *inverse* of a, denoted by a^{-1}.

Definition 3.1.3. An element $a \in R$ is called a *unit* if it has a multiplicative inverse in R.

The set of all units of R is denoted by

$$R^{\times}.$$

3.1.2 Zero Divisors

Definition 3.1.4. Let R be a ring. Nonzero elements $a, b \in R$ are called *zero divisors* if

$$ab = 0.$$

Zero divisors measure the failure of cancellation in a ring.
For example, in $\mathbb{Z}/6\mathbb{Z}$,

$$\overline{2}\,\overline{3} = \overline{0},$$

although neither $\overline{2}$ nor $\overline{3}$ is zero.

3.1.3 Division Rings and Integral Domains

Definition 3.1.5. A ring R with identity is called a *division ring* if every nonzero element is invertible.

A commutative division ring is precisely a field.

Definition 3.1.6. A commutative ring R with identity is called an *integral domain* if it has no nonzero zero divisors.

Thus we have the implications

$$\text{field} \implies \text{integral domain} \implies \text{commutative ring}.$$

The converses are not true in general.

Example 3.1.7. The ring $\mathbb{Z}$ is an integral domain but not a field.

Example 3.1.8. If F is a field, then $F[x]$ is an integral domain.

Example 3.1.9. For $n \geq 2$, the matrix ring $M_n(F)$ is generally noncommutative and contains zero divisors.

3.2 Subrings and Ideals

Definition 3.2.1. Let R be a ring. A subset $S \subseteq R$ is called a *subring* if S is itself a ring under the operations inherited from R.

Example 3.2.2. The ring $\mathbb{Z}$ is a subring of .

Example 3.2.3. The diagonal matrices form a subring of $M_n(F)$.

Subrings are the natural analogue of subgroups, but quotient rings require a stronger condition.

Definition 3.2.4. Let R be a ring. A subring $I \subseteq R$ is called a *two-sided ideal* if

$$rI \subseteq I, \qquad Ir \subseteq I$$

for every $r \in R$.

In a commutative ring, this reduces to

$$rI \subseteq I.$$

We write

$$I \trianglelefteq R$$

to indicate that I is an ideal of R.

Example 3.2.5. For every integer m,

$$m\mathbb{Z}$$

is an ideal of $\mathbb{Z}$.

3.2.1 Generated Ideals

Definition 3.2.6. Let $M \subseteq R$. The *ideal generated by* M is the intersection of all ideals of R containing M.

It is denoted by

$$(M).$$

For a single element $a \in R$, the ideal generated by a is called a *principal ideal* and is denoted by

$$(a).$$

If R is commutative with identity, then

$$(a) = \{ra : r \in R\}.$$

More generally, for

$$M = \{a_1, \ldots, a_n\},$$

we have

$$(a_1, \ldots, a_n) = \{r_1a_1 + \cdots + r_na_n : r_i \in R\}.$$

3.3 Quotient Rings

Let $I \trianglelefteq R$. Consider the additive cosets

$$R/I = \{r + I : r \in R\}.$$

Addition is defined by

$$(r_1 + I) + (r_2 + I) = (r_1 + r_2) + I,$$

and multiplication by

$$(r_1 + I)(r_2 + I) = r_1r_2 + I.$$

Theorem 3.3.1. If $I \trianglelefteq R$, then the operations above are well-defined and make R/I into a ring.

Proof. Suppose

$$r_1 + I = r_1' + I, \qquad r_2 + I = r_2' + I.$$

Then

$$r_1 - r_1' \in I, \qquad r_2 - r_2' \in I.$$

For multiplication,

$$r_1r_2 - r_1'r_2' = r_1(r_2 - r_2') + (r_1 - r_1')r_2'.$$

Since I is an ideal, both terms belong to I. Hence

$$r_1r_2 + I = r_1'r_2' + I.$$

Thus multiplication is well-defined. □

Example 3.3.2. For $m \geq 1$,

$$\mathbb{Z}/m\mathbb{Z}$$

is a quotient ring of $\mathbb{Z}$.

Example 3.3.3. Let F be a field. Then

$$F[x]/(x^2)$$

consists of elements of the form

$$a + bx + (x^2), \qquad a, b \in F.$$

3.4 Ring Homomorphisms

Definition 3.4.1. Let R_1 and R_2 be rings. A map

$$\varphi : R_1 \longrightarrow R_2$$

is a *ring homomorphism* if

$$\varphi(a+b) = \varphi(a) + \varphi(b)$$

and

$$\varphi(ab) = \varphi(a)\varphi(b)$$

for all $a, b \in R_1$.

Its kernel and image are

$$\ker \varphi = \{r \in R_1 : \varphi(r) = 0\},$$

and

$$\operatorname{Im} \varphi = \{\varphi(r) : r \in R_1\}.$$

Lemma 3.4.2. Let

$$\varphi : R_1 \longrightarrow R_2$$

be a ring homomorphism. Then

$$\ker \varphi$$

is an ideal of R_1, and

$$\operatorname{Im} \varphi$$

is a subring of R_2.

Proof. If $a, b \in \ker \varphi$, then

$$\varphi(a-b) = \varphi(a) - \varphi(b) = 0.$$

Hence $a - b \in \ker \varphi$. If $r \in R_1$ and $a \in \ker \varphi$, then

$$\varphi(ra) = \varphi(r)\varphi(a) = 0,$$

and similarly

$$\varphi(ar) = 0.$$

Thus $\ker \varphi$ is an ideal. □

3.4.1 First Isomorphism Theorem

Theorem 3.4.3 (First Isomorphism Theorem for Rings)**.** Let

$$\varphi : R_1 \longrightarrow R_2$$

be a ring homomorphism. Then

$$R_1/\ker \varphi \cong \operatorname{Im} \varphi.$$

The isomorphism is induced by

$$r + \ker \varphi \longmapsto \varphi(r).$$

3.4.2 Correspondence of Ideals

Let $I \trianglelefteq R$, and let

$$\pi : R \longrightarrow R/I$$

be the natural quotient map.

Theorem 3.4.4 (Correspondence Theorem)**.** There is a one-to-one correspondence between the ideals of R containing I and the ideals of R/I.

Explicitly, if

$$I \subseteq J \trianglelefteq R,$$

then the corresponding ideal of R/I is

$$J/I.$$

Theorem 3.4.5. If

$$I \subseteq J \trianglelefteq R,$$

then

$$(R/I)/(J/I) \cong R/J.$$

This is the ring-theoretic version of the third isomorphism theorem.

3.4.3 A Second Isomorphism Theorem

Let $I \trianglelefteq R$ and let $S \leq R$ be a subring. Then

$$I + S$$

is a subring of R, while

$$I \cap S$$

is an ideal of S.
Moreover,

$$\frac{I+S}{I} \cong \frac{S}{I \cap S}.$$

3.5 Direct Products of Rings

Let R and S be rings. Define

$$R \times S = \{(r, s) : r \in R,\ s \in S\}$$

with

$$(r_1, s_1) + (r_2, s_2) = (r_1 + r_2, s_1 + s_2),$$

and

$$(r_1, s_1)(r_2, s_2) = (r_1 r_2, s_1 s_2).$$

Then $R \times S$ is a ring.
If both rings have identities, then

$$1_{R\times S} = (1_R, 1_S).$$

The subsets

$$R \times \{0\}, \qquad \{0\} \times S$$

are ideals of $R \times S$.

3.6 Prime and Maximal Ideals

From this point, assume that R is a commutative ring with identity.

Definition 3.6.1. A proper ideal $P \trianglelefteq R$ is called a *prime ideal* if

$$ab \in P$$

implies

$$a \in P \qquad \text{or} \qquad b \in P.$$

Definition 3.6.2. A proper ideal $M \trianglelefteq R$ is called a *maximal ideal* if there is no ideal I satisfying

$$M \subsetneq I \subsetneq R.$$

These notions can be characterized entirely through quotient rings.

Theorem 3.6.3. Let $I \trianglelefteq R$. Then

$$I \text{ is prime} \quad \Longleftrightarrow \quad R/I \text{ is an integral domain.}$$

Proof. Suppose I is prime. If

$$(a+I)(b+I) = I,$$

then $ab \in I$. Hence $a \in I$ or $b \in I$, so

$$a + I = I \qquad \text{or} \qquad b + I = I.$$

Thus R/I has no nonzero zero divisors.
Conversely, suppose R/I is an integral domain. If $ab \in I$, then

$$(a+I)(b+I) = I.$$

Hence

$$a + I = I \qquad \text{or} \qquad b + I = I,$$

so $a \in I$ or $b \in I$. □

Theorem 3.6.4. Let $I \trianglelefteq R$. Then

$$I \text{ is maximal} \quad \Longleftrightarrow \quad R/I \text{ is a field.}$$

Proof. Suppose I is maximal. Let

$$a + I \neq I.$$

Then $a \notin I$, so

$$I + (a)$$

is an ideal strictly containing I. By maximality,

$$I + (a) = R.$$

Hence there exist $r \in R$ and $i \in I$ such that

$$ra + i = 1.$$

Therefore

$$(r + I)(a + I) = 1 + I,$$

so every nonzero element of R/I is invertible.
Conversely, suppose R/I is a field. If

$$I \subseteq J \trianglelefteq R,$$

then J/I is an ideal of R/I. A field has only the ideals $\{0\}$ and itself. Hence

$$J = I \qquad \text{or} \qquad J = R.$$

Thus I is maximal. □

Corollary 3.6.5. Every maximal ideal in a commutative ring with identity is prime.

Indeed,

$$R/I \text{ field} \Longrightarrow R/I \text{ integral domain.}$$

3.7 Chinese Remainder Theorem

The Chinese Remainder Theorem may be interpreted as a statement about quotient rings.

Definition 3.7.1. Two ideals $I, J \trianglelefteq R$ are called *coprime* if

$$I + J = R.$$

Lemma 3.7.2. If I_1, I_2 are both coprime to J, then

$$I_1 I_2$$

is also coprime to J.

Proof. Since

$$I_1 + J = R, \qquad I_2 + J = R,$$

there exist

$$a_1 \in I_1, \quad a_2 \in I_2, \quad b_1, b_2 \in J$$

such that

$$a_1 + b_1 = 1, \qquad a_2 + b_2 = 1.$$

Multiplying gives

$$1 = a_1 a_2 + a_1 b_2 + b_1 a_2 + b_1 b_2.$$

The first term belongs to $I_1 I_2$, while the remaining terms belong to J. Therefore

$$I_1 I_2 + J = R.$$

□

Theorem 3.7.3 (Chinese Remainder Theorem)**.** Let R be a commutative ring with identity, and let

$$I_1, \ldots, I_n$$

be pairwise coprime ideals. Then

$$R/(I_1 I_2 \cdots I_n) \cong R/I_1 \times \cdots \times R/I_n.$$

Consider the homomorphism

$$\varphi : R \longrightarrow R/I_1 \times \cdots \times R/I_n$$

defined by

$$\varphi(r) = (r + I_1, \ldots, r + I_n).$$

Its kernel is

$$I_1 \cap \cdots \cap I_n.$$

For pairwise coprime ideals,

$$I_1 \cap \cdots \cap I_n = I_1 I_2 \cdots I_n,$$

and the coprimality condition makes φ surjective. The result follows from the first isomorphism theorem.

Corollary 3.7.4. If

$$n = m_1 m_2 \cdots m_r$$

and the m_i are pairwise coprime, then

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/m_1\mathbb{Z} \times \cdots \times \mathbb{Z}/m_r\mathbb{Z}.$$

3.8 Fraction Fields and Localization

Let R be an integral domain.
The construction of from $\mathbb{Z}$ can be generalized to every integral domain.
Consider pairs

$$(a, b), \qquad a, b \in R, \quad b \neq 0,$$

with the equivalence relation

$$(a, b) \sim (c, d) \quad \Longleftrightarrow \quad ad = bc.$$

Denote the equivalence class of (a, b) by

$$\frac{a}{b}.$$

Define

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd},$$

and

$$\frac{a}{b}\frac{c}{d} = \frac{ac}{bd}.$$

Theorem 3.8.1. The set of equivalence classes under these operations forms a field, called the *fraction field* of R.

It is commonly denoted by

$$\mathrm{Frac}(R).$$

For example,

$$\mathrm{Frac}(\mathbb{Z}) = .$$

3.8.1 Localization

A related construction allows only selected denominators.
Let R be a commutative ring with identity, and let $S \subseteq R$ be a multiplicatively closed subset containing no zero divisor.
One forms fractions

$$\frac{a}{s}, \qquad a \in R, \quad s \in S.$$

The resulting ring is denoted by

$$S^{-1}R$$

and is called the *localization* of R at S.
The fraction field is the special case in which R is an integral domain and

$$S = R \setminus \{0\}.$$

3.9 Divisibility in Integral Domains

From this point onward, let D be an integral domain.

Definition 3.9.1. For $a, b \in D$, we say that b *divides* a, written

$$b \mid a,$$

if there exists $c \in D$ such that

$$a = bc.$$

If $a = bc$, then b is called a factor of a, and a is called a multiple of b.

3.9.1 Associates

Definition 3.9.2. Two elements $a, b \in D$ are called *associates* if

$$a = ub$$

for some unit $u \in D^\times$.

Equivalently,

$$(a) = (b).$$

Thus associates represent essentially the same factor.

3.10 Irreducible and Prime Elements

Definition 3.10.1. A nonzero nonunit $p \in D$ is called *irreducible* if

$$p = ab$$

implies that either a or b is a unit.

Definition 3.10.2. A nonzero nonunit $p \in D$ is called a *prime element* if

$$p \mid ab$$

implies

$$p \mid a \qquad \text{or} \qquad p \mid b.$$

These two notions are related but are not equivalent in every integral domain.

Proposition 3.10.3. In an integral domain, every prime element is irreducible.

Proof. Let p be prime and suppose

$$p = ab.$$

Then $p \mid ab$, so $p \mid a$ or $p \mid b$. Suppose $p \mid a$. Then $a = pc$ for some $c \in D$, and therefore

$$p = ab = pcb.$$

Since D is an integral domain and $p \neq 0$,

$$cb = 1.$$

Hence b is a unit. Therefore p is irreducible. □

The converse need not hold.

Example 3.10.4. In the integral domain

$$\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\},$$

the element 2 is irreducible but not prime.

Indeed,

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}),$$

showing the failure of unique factorization.

3.11 Unique Factorization Domains

Definition 3.11.1. An integral domain D is called a *unique factorization domain*, abbreviated *UFD*, if:

1. every nonzero nonunit element of D can be written as a finite product of irreducible elements;
2. this factorization is unique up to order and multiplication of the irreducible factors by units.

Thus if

$$a = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$$

are two irreducible factorizations, then

$$r = s$$

and, after rearranging,

$$p_i \sim q_i$$

for every i.

Theorem 3.11.2. Let D be an integral domain. Then D is a UFD if and only if:

1. every nonzero nonunit can be written as a finite product of irreducible elements;
2. every irreducible element is prime.

Proof. Suppose first that D is a UFD. Let p be irreducible and suppose

$$p \mid ab.$$

Then

$$ab = pc$$

for some $c \in D$. Factor a, b, c into irreducibles. By uniqueness of factorization, p must be associate to one of the irreducible factors appearing in a or b. Hence

$$p \mid a \qquad \text{or} \qquad p \mid b.$$

Thus p is prime.
Conversely, suppose factorizations into irreducibles exist and every irreducible is prime. Let

$$p_1 \cdots p_r = q_1 \cdots q_s.$$

Since p_1 is prime,

$$p_1 \mid q_1 \cdots q_s.$$

Hence $p_1 \mid q_j$ for some j. Since q_j is irreducible, p_1 and q_j are associates. Cancelling them and repeating the argument proves uniqueness. □

3.12 Principal Ideal Domains

Definition 3.12.1. An integral domain D is called a *principal ideal domain*, abbreviated *PID*, if every ideal of D is principal.

Thus every ideal has the form

$$I = (a)$$

for some $a \in D$.

Example 3.12.2. The ring $\mathbb{Z}$ is a PID.

Example 3.12.3. If F is a field, then $F[x]$ is a PID.

3.12.1 Prime and Irreducible Elements in a PID

Proposition 3.12.4. Let D be a PID and let $p \in D$ be a nonzero nonunit. Then the following statements are equivalent:

1. p is irreducible;
2. p is prime;
3. (p) is a prime ideal;
4. (p) is a maximal ideal.

Proof. Suppose p is irreducible. Let I be an ideal satisfying

$$(p) \subseteq I \subseteq D.$$

Since D is a PID,

$$I = (a)$$

for some $a \in D$. Since $p \in (a)$,

$$p = ab$$

for some $b \in D$. Irreducibility of p implies that a or b is a unit.
If a is a unit, then $I = D$. If b is a unit, then

$$(a) = (p).$$

Therefore (p) is maximal.
A maximal ideal is prime, so (p) is prime. Hence p is a prime element.
Finally, every prime element in an integral domain is irreducible. □

3.12.2 PID Implies UFD

Theorem 3.12.5. Every PID is a UFD.

Proof. By the previous proposition, every irreducible element in a PID is prime. It remains to prove existence of irreducible factorizations.
Suppose some nonzero nonunit a_1 cannot be expressed as a finite product of irreducibles. Then a_1 is not irreducible, so

$$a_1 = a_2 b_2$$

with both a_2 and b_2 nonunits. Hence

$$(a_1) \subsetneq (a_2).$$

Repeating this process gives a strictly increasing chain

$$(a_1) \subsetneq (a_2) \subsetneq (a_3) \subsetneq \cdots .$$

Let

$$I = \bigcup_{n=1}^{\infty} (a_n).$$

Then I is an ideal. Since D is a PID,

$$I = (b)$$

for some $b \in D$.
Since $b \in I$, there exists N such that

$$b \in (a_N).$$

Therefore

$$I = (b) \subseteq (a_N).$$

But $(a_N) \subseteq I$, so

$$I = (a_N),$$

contradicting

$$(a_N) \subsetneq (a_{N+1}) \subseteq I.$$

Thus every nonzero nonunit has an irreducible factorization. □

Therefore

$$\boxed{\text{PID} \Longrightarrow \text{UFD}}.$$

The converse is not true in general.

3.13 Euclidean Domains

A Euclidean domain is an integral domain in which a division algorithm is available.

Definition 3.13.1. Let D be an integral domain. A function

$$\nu : D \setminus \{0\} \longrightarrow \mathbb{N}$$

is called a *Euclidean valuation* if for every $a, b \in D$ with $b \neq 0$, there exist $q, r \in D$ such that

$$a = qb + r$$

where either $r = 0$ or

$$\nu(r) < \nu(b).$$

Definition 3.13.2. An integral domain admitting a Euclidean valuation is called a *Euclidean domain*, abbreviated *ED*.

Example 3.13.3. The ring $\mathbb{Z}$ is Euclidean with

$$\nu(n) = |n|.$$

Example 3.13.4. If F is a field, then $F[x]$ is Euclidean with

$$\nu(f) = \deg f.$$

3.13.1 Euclidean Domains Are PIDs

Theorem 3.13.5. Every Euclidean domain is a PID.

Proof. Let D be Euclidean and let

$$I \neq \{0\}$$

be an ideal of D. Choose $b \in I \setminus \{0\}$ such that

$$\nu(b)$$

is minimal among all nonzero elements of I.
Clearly,

$$(b) \subseteq I.$$

Let $a \in I$. By the Euclidean property,

$$a = qb + r,$$

where either $r = 0$ or

$$\nu(r) < \nu(b).$$

Since

$$r = a - qb \in I,$$

minimality of $\nu(b)$ forces $r = 0$. Hence

$$a = qb \in (b).$$

Thus

$$I = (b).$$

Therefore D is a PID. □

Combining the results above,

$$\boxed{\text{ED} \Longrightarrow \text{PID} \Longrightarrow \text{UFD} \Longrightarrow \text{ID}}.$$

None of the reverse implications holds in general.

3.13.2 Gaussian Integers

The Gaussian integers are

$$\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}.$$

Define

$$N(a + bi) = a^2 + b^2.$$

This norm can be used as a Euclidean valuation. Hence

$$\mathbb{Z}[i]$$

is a Euclidean domain and therefore also a PID and a UFD.

3.14 Polynomial Rings over UFDs

We now study factorization in

$$R[x]$$

when R is a UFD.

3.14.1 Content and Primitive Polynomials

Let

$$f(x) = a_0 + a_1x + \cdots + a_nx^n \in R[x].$$

Definition 3.14.1. The greatest common divisor of the coefficients of f, defined up to multiplication by a unit, is called the *content* of f, denoted by

$$c(f).$$

Definition 3.14.2. A polynomial $f \in R[x]$ is called *primitive* if

$$c(f)$$

is a unit.

Every nonzero polynomial may therefore be written in the form

$$f = c(f)f_0,$$

where f_0 is primitive.

3.14.2 Gauss's Lemma

Lemma 3.14.3 (Gauss's Lemma)**.** Let R be a UFD and let $f, g \in R[x]$. Then

$$c(fg) \sim c(f)c(g).$$

In particular, if f and g are primitive, then fg is primitive.

Proof. It suffices to prove the primitive case.
Suppose f and g are primitive but fg is not. Then there exists a prime $p \in R$ dividing every coefficient of fg.
Write

$$f(x) = a_0 + a_1x + \cdots + a_nx^n,$$

and

$$g(x) = b_0 + b_1x + \cdots + b_mx^m.$$

Since f and g are primitive, there exist indices i, j such that

$$p \nmid a_i, \qquad p \nmid b_j,$$

while p divides the preceding relevant coefficients.
Consider the coefficient of x^{i+j} in fg:

$$c_{i+j} = a_0 b_{i+j} + \cdots + a_i b_j + \cdots + a_{i+j} b_0.$$

All terms except $a_i b_j$ are divisible by p. Since p is prime and

$$p \nmid a_i, \qquad p \nmid b_j,$$

we have

$$p \nmid a_i b_j.$$

Thus $p \nmid c_{i+j}$, a contradiction. □

3.14.3 Irreducibility over the Fraction Field

Let $K = \mathrm{Frac}(R)$.

Theorem 3.14.4. Let R be a UFD and let $f \in R[x]$ be primitive. Then

$$f \text{ is irreducible in } R[x]$$

if and only if

$$f \text{ is irreducible in } K[x].$$

This allows questions about irreducibility over an integral domain to be transferred to its fraction field.

3.14.4 Polynomial Rings Preserve Unique Factorization

Theorem 3.14.5. If R is a UFD, then

$$R[x]$$

is also a UFD.

Proof. Let

$$K = \mathrm{Frac}(R).$$

Since K is a field,

$$K[x]$$

is a Euclidean domain and hence a UFD.
Gauss's lemma allows irreducible factorizations in $K[x]$ to be cleared of denominators and converted into factorizations in $R[x]$. Primitive factors remain primitive, and uniqueness in $K[x]$, together with uniqueness of factorization of contents in R, gives uniqueness in $R[x]$.
Therefore $R[x]$ is a UFD. □

By induction,

$$R[x_1, \ldots, x_n]$$

is a UFD whenever R is a UFD.
In particular,

$$\mathbb{Z}[x_1, \ldots, x_n]$$

is a UFD.

3.15 The Hierarchy of Factorization Domains

The main classes of integral domains introduced in this chapter fit into the hierarchy

$$\boxed{\text{Euclidean Domain} \Longrightarrow \text{Principal Ideal Domain} \Longrightarrow \text{Unique Factorization Domain} \Longrightarrow \text{Integral Domai}}$$

The meaning of each step is different.
A Euclidean domain has a division algorithm. A principal ideal domain requires every ideal to be generated by one element. A unique factorization domain requires the existence and uniqueness of factorizations into irreducibles. An integral domain requires only the absence of zero divisors.
Thus the implications may be interpreted as

$$\text{division algorithm} \Longrightarrow \text{simple ideal structure} \Longrightarrow \text{unique factorization} \Longrightarrow \text{cancellation.}$$

This hierarchy provides the main structural framework for factorization in commutative ring theory and will also be used in the study of polynomial rings and field extensions in the next chapter.

Chapter 4

Field Theory

4.1 Fields, Subfields, and Characteristic

We now turn from rings to fields. Recall that a field is a commutative ring in which every nonzero element is invertible.

Definition 4.1.1. Let E be a field. A subset $F \subseteq E$ is called a *subfield* of E if F is itself a field under the operations inherited from E.

If $F \subseteq E$, we also say that E is an *extension field* of F, and write

$$E/F.$$

Typical examples are

$$\subseteq \mathbb{R} \subseteq \mathbb{C}.$$

4.1.1 Characteristic

Definition 4.1.2. Let F be a field. The *characteristic* of F is the smallest positive integer n such that

$$n \cdot 1_F = 0,$$

if such an integer exists. If no such positive integer exists, we define

$$\text{char}(F) = 0.$$

Lemma 4.1.3. The characteristic of a field is either 0 or a prime number.

Proof. Suppose

$$\text{char}(F) = n > 0$$

and that $n = ab$ with $1 < a, b < n$. Then

$$(a \cdot 1_F)(b \cdot 1_F) = n \cdot 1_F = 0.$$

Since F has no zero divisors,

$$a \cdot 1_F = 0 \qquad \text{or} \qquad b \cdot 1_F = 0,$$

contradicting the minimality of n. Hence n is prime. □

4.1.2 Prime Fields

Theorem 4.1.4. Every field contains a smallest subfield, which is isomorphic to either

or

$$\mathbb{F}_p$$

for some prime p.

Definition 4.1.5. The fields

$$\text{and} \qquad \mathbb{F}_p$$

are called *prime fields*.

If $\text{char}(F) = 0$, then the prime subfield of F is isomorphic to . If $\text{char}(F) = p$, then it is isomorphic to $\mathbb{F}_p$.

4.2 Field Extensions

Let $F \subseteq E$ be fields.

Definition 4.2.1. If $S \subseteq E$, then $F(S)$ denotes the intersection of all subfields of E containing both F and S.

Thus $F(S)$ is the smallest subfield of E containing F and S.
If

$$S = \{\alpha\},$$

we write

$$F(\alpha).$$

If

$$S = \{\alpha_1, \ldots, \alpha_n\},$$

we write

$$F(\alpha_1, \ldots, \alpha_n).$$

Definition 4.2.2. An extension E/F is called a *simple extension* if

$$E = F(\alpha)$$

for some $\alpha \in E$.

Example 4.2.3. We have

$$(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in\}.$$

This is a simple extension of .

4.3 Algebraic and Transcendental Elements

Definition 4.3.1. Let E/F be a field extension and let $\alpha \in E$. The element α is called *algebraic over* F if there exists a nonzero polynomial $f(x) \in F[x]$ such that

$$f(\alpha) = 0.$$

Otherwise, α is called *transcendental over* F.

Example 4.3.2. The element $\sqrt{2}$ is algebraic over , since

$$(\sqrt{2})^2 - 2 = 0.$$

4.3.1 Evaluation Homomorphism

For $\alpha \in E$, define

$$\mathrm{ev}_\alpha : F[x] \longrightarrow E$$

by

$$f(x) \longmapsto f(\alpha).$$

This is a ring homomorphism.
If α is transcendental, then

$$\ker(\mathrm{ev}_\alpha) = \{0\}.$$

Hence $F[\alpha]$ behaves like a polynomial ring.
If α is algebraic, the kernel is nontrivial.

4.4 Minimal Polynomials and Simple Extensions

Definition 4.4.1. Let α be algebraic over F. The unique monic polynomial of smallest positive degree satisfying

$$m_{\alpha,F}(\alpha) = 0$$

is called the *minimal polynomial* of α over F.

We often write simply

$$m_\alpha(x)$$

when the base field is clear.

Proposition 4.4.2. The minimal polynomial $m_\alpha(x)$ is irreducible over F. Moreover, if $f(x) \in F[x]$ satisfies $f(\alpha) = 0$, then

$$m_\alpha(x) \mid f(x).$$

Proof. Suppose

$$m_\alpha(x) = g(x)h(x)$$

with both g and h of positive degree. Then

$$0 = m_\alpha(\alpha) = g(\alpha)h(\alpha).$$

Since E is a field,

$$g(\alpha) = 0 \qquad \text{or} \qquad h(\alpha) = 0,$$

contradicting minimality of the degree of m_α. Hence m_α is irreducible.
For the divisibility statement, divide f by m_α:

$$f(x) = q(x)m_\alpha(x) + r(x), \qquad \deg r < \deg m_\alpha.$$

Evaluating at α gives $r(\alpha) = 0$. By minimality, $r = 0$. □

Theorem 4.4.3. If α is algebraic over F, then

$$F(\alpha) \cong F[x]/(m_\alpha(x)).$$

Proof. Consider

$$\mathrm{ev}_\alpha : F[x] \longrightarrow F(\alpha).$$

Its kernel is

$$(m_\alpha(x)).$$

Since m_α is irreducible, the quotient

$$F[x]/(m_\alpha)$$

is a field. The first isomorphism theorem gives

$$F[x]/(m_\alpha) \cong F[\alpha] = F(\alpha).$$

□

4.4.1 Adjoining Roots

Theorem 4.4.4. Let $f(x) \in F[x]$ be irreducible. Then there exists an extension field E/F in which f has a root.

Proof. Since $f(x)$ is irreducible,

$$(f(x))$$

is a maximal ideal of $F[x]$. Hence

$$E = F[x]/(f(x))$$

is a field.
Let

$$\alpha = x + (f(x)).$$

Then

$$f(\alpha) = 0$$

in E. □

This construction is one of the fundamental uses of quotient rings in field theory.

4.5 Degree of a Field Extension

Let E/F be a field extension. Since E is naturally a vector space over F, we may consider its dimension.

Definition 4.5.1. The *degree* of E/F is

$$[E : F] = \dim_F E.$$

If

$$[E : F] < \infty,$$

the extension is called a *finite extension*.

Proposition 4.5.2. If α is algebraic over F, then

$$[F(\alpha) : F] = \deg m_\alpha.$$

Indeed, if

$$\deg m_\alpha = n,$$

then

$$1, \alpha, \alpha^2, \ldots, \alpha^{n-1}$$

form an F-basis of $F(\alpha)$.

4.5.1 Tower Law

Theorem 4.5.3 (Tower Law)**.** Let

$$F \subseteq E \subseteq K$$

be fields. If

$$[K : E] = m \qquad \text{and} \qquad [E : F] = n,$$

then

$$[K : F] = mn.$$

Proof. Let

$$\{\alpha_1, \ldots, \alpha_m\}$$

be a basis of K over E, and let

$$\{\beta_1, \ldots, \beta_n\}$$

be a basis of E over F.
Then

$$\{\alpha_i \beta_j : 1 \leq i \leq m,\ 1 \leq j \leq n\}$$

is a basis of K over F. Hence

$$[K : F] = mn.$$

□

Example 4.5.4. Consider

$$\subseteq (\sqrt{2}) \subseteq (\sqrt{2}, \sqrt{3}).$$

We have

$$[(\sqrt{2}) :] = 2$$

and

$$[(\sqrt{2}, \sqrt{3}) : (\sqrt{2})] = 2.$$

Hence

$$[(\sqrt{2}, \sqrt{3}) :] = 4.$$

4.5.2 Constructibility Example

The degree of a field extension can be used to study geometric constructibility.
For example,

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta.$$

Taking $\theta = 20°$, we obtain

$$4x^3 - 3x - \frac{1}{2} = 0, \qquad x = \cos 20°.$$

Equivalently,

$$8x^3 - 6x - 1 = 0.$$

If this polynomial is irreducible over , then

$$[(x) :] = 3.$$

Since a classical straightedge-and-compass construction produces extensions of degree a power of 2, this gives the algebraic obstruction behind the impossibility of trisecting a general angle.

4.6 Algebraic Extensions

Definition 4.6.1. An extension E/F is called *algebraic* if every element of E is algebraic over F.

Proposition 4.6.2. Every finite extension is algebraic.

Proof. Let

$$[E : F] = n < \infty$$

and let $\alpha \in E$. The $n + 1$ elements

$$1, \alpha, \alpha^2, \ldots, \alpha^n$$

are linearly dependent over F. Hence there exist $a_0, \ldots, a_n \in F$, not all zero, such that

$$a_0 + a_1\alpha + \cdots + a_n\alpha^n = 0.$$

Thus α is algebraic over F. □

Definition 4.6.3. A field F is called *algebraically closed* if every nonconstant polynomial in $F[x]$ has a root in F.

Equivalently, every nonconstant polynomial over F splits into linear factors over F.

4.7 Finite Fields

Let F be a finite field.

Theorem 4.7.1. There exist a prime p and a positive integer n such that

$$|F| = p^n.$$

Proof. Let

$$\mathrm{char}(F) = p.$$

Then F contains $\mathbb{F}_p$, and F is a finite-dimensional vector space over $\mathbb{F}_p$. If

$$[F : \mathbb{F}_p] = n,$$

then

$$|F| = p^n.$$

□

The additive group of F is therefore an elementary abelian p-group.

4.7.1 The Multiplicative Group

Theorem 4.7.2. The multiplicative group

$$F^\times = F \setminus \{0\}$$

of a finite field is cyclic.

Thus, if

$$|F| = q = p^n,$$

then

$$F^\times \cong \mathbb{Z}_{q-1}.$$

A finite field of order q is commonly denoted by

$$\mathbb{F}_q$$

or

$$GF(q).$$

4.7.2 Frobenius Automorphism

Suppose

$$\mathrm{char}(F) = p.$$

Define

$$\mathrm{Fr} : F \longrightarrow F$$

by

$$\mathrm{Fr}(x) = x^p.$$

Since

$$(x+y)^p = x^p + y^p$$

in characteristic p, and

$$(xy)^p = x^p y^p,$$

the Frobenius map is a field homomorphism.
For a finite field, it is bijective and therefore an automorphism.

4.8 Primitive Element Theorem

A finite extension need not obviously be generated by a single element. Under the assumptions used in the notes, however, this can often be achieved.

Theorem 4.8.1 (Primitive Element Theorem)**.** Assume

$$\mathrm{char}(F) = 0.$$

Every finite extension E/F is simple. That is, there exists $\alpha \in E$ such that

$$E = F(\alpha).$$

The essential case is

$$E = F(\alpha, \beta).$$

Choose

$$\gamma = \alpha + c\beta$$

for a suitable $c \in F$. One chooses c so that the conjugates of α and β do not create unwanted coincidences. Then one shows that

$$\beta \in F(\gamma),$$

and hence

$$\alpha = \gamma - c\beta \in F(\gamma).$$

Therefore

$$F(\alpha, \beta) = F(\gamma).$$

Repeating this argument gives the finite-generator case.

Proposition 4.8.2. Every finite extension of a finite field is simple.

Indeed, the multiplicative group of a finite field is cyclic. If E/F is finite and α generates $E^\times$, then

$$E = F(\alpha).$$

4.9 Splitting Fields

Definition 4.9.1. Let E/F be a field extension and let

$$f(x) \in F[x].$$

A *splitting field* of f over F is the smallest subfield $K \subseteq E$ containing F and all roots of f.

Equivalently, if

$$\alpha_1, \ldots, \alpha_n$$

are all roots of f, then

$$K = F(\alpha_1, \ldots, \alpha_n).$$

Example 4.9.2. Consider

$$f(x) = x^3 - 2$$

over . Let

$$\alpha = \sqrt[3]{2}$$

and let

$$\omega = e^{2\pi i/3}.$$

The roots are

$$\alpha, \qquad \omega\alpha, \qquad \omega^2\alpha.$$

Thus the splitting field is

$$(\alpha, \omega).$$

The field

$$(\alpha)$$

is not the splitting field, since it does not contain the other two roots.

Proposition 4.9.3. Any two splitting fields of the same polynomial over F are isomorphic by an isomorphism fixing F.

4.9.1 Roots of an Irreducible Polynomial

Under the characteristic-zero assumption used in the notes, roots of an irreducible polynomial behave symmetrically inside a splitting field.

Theorem 4.9.4. Assume $\mathrm{char}(F) = 0$. Let K/F be a splitting field extension. If an irreducible polynomial

$$f(x) \in F[x]$$

has one root in K, then all its roots lie in K.

This property motivates the notion of a normal extension.

4.10 Normal Extensions

Definition 4.10.1. An algebraic extension E/F is called *normal* if every irreducible polynomial $f(x) \in F[x]$ having one root in E splits completely over E.

Under the setting of the notes, a splitting field of a polynomial over F is a normal extension of F.
Different polynomials may have the same splitting field and therefore lead to the same Galois group.

4.11 Field Automorphisms and Galois Groups

Definition 4.11.1. Let E/F be a field extension. An automorphism

$$\sigma : E \longrightarrow E$$

is called an *F-automorphism* if

$$\sigma(a) = a$$

for every $a \in F$.

Definition 4.11.2. The set of all F-automorphisms of E, under composition, is called the *Galois group* of E/F, denoted by

$$\mathrm{Gal}(E/F).$$

If E is the splitting field of $f(x) \in F[x]$, then $\mathrm{Gal}(E/F)$ is also called the Galois group of f, and is sometimes written

$$\mathrm{Gal}(f).$$

4.11.1 Automorphisms Permute Roots

Let

$$f(x) \in F[x]$$

and

$$\sigma \in \mathrm{Gal}(E/F).$$

If $f(\alpha) = 0$, then

$$f(\sigma(\alpha)) = \sigma(f(\alpha)) = 0.$$

Thus every F-automorphism permutes the roots of f.
Consequently, if f has n distinct roots in its splitting field, then

$$\mathrm{Gal}(f)$$

can be viewed as a subgroup of S_n.

Example 4.11.3. Consider

$$\mathbb{C}/\mathbb{R}.$$

Complex conjugation

$$\sigma(a + bi) = a - bi$$

fixes every real number. Hence

$$\mathrm{Gal}(\mathbb{C}/\mathbb{R}) \cong \mathbb{Z}_2.$$

Example 4.11.4. Let

$$E = (\sqrt{2}).$$

The nontrivial automorphism is

$$\sigma(a + b\sqrt{2}) = a - b\sqrt{2}.$$

Hence

$$\mathrm{Gal}(E/) \cong \mathbb{Z}_2.$$

4.11.2 The Splitting Field of $x^3 - 2$

Let

$$\alpha = \sqrt[3]{2}, \qquad \omega = e^{2\pi i/3},$$

and

$$E = (\alpha, \omega).$$

Then E is the splitting field of $x^3 - 2$ over .
Every automorphism permutes the three roots

$$\alpha, \quad \omega\alpha, \quad \omega^2\alpha.$$

The resulting Galois group is

$$\mathrm{Gal}(E/) \cong S_3.$$

The intermediate field

$$(\omega)$$

has degree 2 over , while

$$E/(\omega)$$

has degree 3.
This example already exhibits the basic relationship between intermediate fields and subgroups of a Galois group.

4.12 Fixed Fields and the Galois Correspondence

Let E/F be a finite normal extension in the characteristic-zero setting of these notes.
For a subgroup

$$H \leq \mathrm{Gal}(E/F),$$

define its *fixed field* by

$$E^H = \{x \in E : \sigma(x) = x \text{ for every } \sigma \in H\}.$$

Conversely, for an intermediate field

$$F \subseteq L \subseteq E,$$

consider

$$\mathrm{Gal}(E/L) = \{\sigma \in \mathrm{Gal}(E/F) : \sigma(x) = x \text{ for all } x \in L\}.$$

Thus we obtain two operations:

$$L \longmapsto \mathrm{Gal}(E/L),$$

and

$$H \longmapsto E^H.$$

Theorem 4.12.1 (Fundamental Theorem of Galois Theory)**.** Let E/F be a finite Galois extension. There is an inclusion-reversing bijection between intermediate fields

$$F \subseteq L \subseteq E$$

and subgroups

$$H \leq \mathrm{Gal}(E/F),$$

given by

$$L \longmapsto \mathrm{Gal}(E/L)$$

and

$$H \longmapsto E^H.$$

The correspondence reverses inclusion:

$$L_1 \subseteq L_2 \quad \Longleftrightarrow \quad \mathrm{Gal}(E/L_2) \leq \mathrm{Gal}(E/L_1).$$

Moreover,

$$[E : L] = |\mathrm{Gal}(E/L)|$$

and

$$[L : F] = [\mathrm{Gal}(E/F) : \mathrm{Gal}(E/L)].$$

4.12.1 Normal Intermediate Extensions

Theorem 4.12.2. Let

$$F \subseteq L \subseteq E$$

with E/F finite Galois. Then

$$L/F \text{ is normal}$$

if and only if

$$\mathrm{Gal}(E/L) \trianglelefteq \mathrm{Gal}(E/F).$$

When this happens,

$$\mathrm{Gal}(L/F) \cong \frac{\mathrm{Gal}(E/F)}{\mathrm{Gal}(E/L)}.$$

Thus normal subgroups in group theory correspond to normal field extensions in field theory.

4.13 Irreducibility and Transitive Galois Groups

Let $f(x) \in F[x]$ be irreducible and let E be its splitting field.
The Galois group acts on the set of roots of f.

Proposition 4.13.1. If f is irreducible, then

$$\mathrm{Gal}(f)$$

acts transitively on the roots of f.

Thus irreducibility of a polynomial translates into transitivity of a permutation group.

Lemma 4.13.2. Let $G \leq S_p$ act transitively on a set of prime size p. If G contains a transposition, then

$$G = S_p.$$

Proposition 4.13.3. Let

$$f(x) \in [x]$$

be irreducible of prime degree p. Suppose f has exactly two nonreal roots. Then

$$\mathrm{Gal}(f) \cong S_p.$$

Proof. Since f is irreducible, the Galois group acts transitively on its p roots.
Complex conjugation fixes every real root and interchanges the two nonreal roots. Hence it acts as a transposition.
By the previous lemma,

$$\mathrm{Gal}(f) = S_p.$$

□

4.14 Radical Extensions

The original motivation for Galois theory is the question of whether the roots of a polynomial can be expressed using arithmetic operations and radicals.

Definition 4.14.1. Let

$$F = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_r = E$$

be a tower of fields such that

$$F_i = F_{i-1}(\alpha_i)$$

and

$$\alpha_i^{p_i} \in F_{i-1}$$

for some prime p_i.
Such a tower is called a *radical tower*, and E/F is called a *radical extension*.

Definition 4.14.2. A polynomial

$$f(x) \in F[x]$$

is called *solvable by radicals* if its splitting field is contained in some radical extension of F.

For example, the roots of

$$x^n - a$$

are obtained by adjoining an n-th root of a together with appropriate roots of unity.

4.15 Solvability by Radicals and Solvable Groups

This is the final connection between the group theory developed in Chapter 2 and the field theory developed here.
Recall that a group G is solvable if its derived series eventually reaches the identity.
Under the characteristic-zero assumptions of the notes, the central result is the following.

Theorem 4.15.1 (Galois Criterion for Solvability by Radicals)**.** Let

$$f(x) \in F[x], \qquad \mathrm{char}(F) = 0.$$

Then f is solvable by radicals if and only if

$$\mathrm{Gal}(f)$$

is a solvable group.

The proof proceeds in two directions.

4.15.1 Radicals Imply a Solvable Galois Group

Suppose the splitting field E of f is contained in a radical extension

$$F = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_r.$$

After adjoining the necessary roots of unity, the tower may be enlarged so that the relevant extensions are normal.
Each step then corresponds to a cyclic Galois group of prime order:

$$\mathrm{Gal}(F_i/F_{i-1}) \cong \mathbb{Z}_{p_i}.$$

Consequently, the total Galois group admits a normal series whose factor groups are abelian. Hence it is solvable.
Therefore

$$f \text{ solvable by radicals} \Longrightarrow \mathrm{Gal}(f) \text{ solvable}.$$

4.15.2 A Solvable Galois Group Gives Radicals

Conversely, suppose

$$G = \mathrm{Gal}(f)$$

is solvable.
Then there exists a chain

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_r = \{e\},$$

which may be refined so that the factors

$$G_i/G_{i+1}$$

are cyclic groups of prime order.
By the Galois correspondence, this chain gives a tower of fields

$$F = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_r = E.$$

After adjoining the necessary roots of unity, cyclic extensions of prime degree can be obtained by adjoining radicals. Hence the splitting field lies inside a radical extension.
Therefore

$$\mathrm{Gal}(f) \text{ solvable} \Longrightarrow f \text{ solvable by radicals}.$$

Combining the two directions gives

$$\boxed{f \text{ is solvable by radicals} \quad \Longleftrightarrow \quad \mathrm{Gal}(f) \text{ is solvable}}.$$

4.15.3 Why the General Quintic Is Different

The symmetric group

$$S_5$$

is not solvable, because

$$A_5$$

is a non-abelian simple group.
Therefore, if a quintic polynomial has Galois group S_5, it cannot be solved by radicals.
The notes give the example

$$f(x) = x^5 - 8x + 2 \in [x],$$

whose Galois group is identified with S_5. Hence this polynomial is not solvable by radicals.
This provides the conceptual explanation for the failure of a general formula by radicals for quintic equations:

$$\boxed{\text{polynomial} \longrightarrow \text{splitting field} \longrightarrow \text{Galois group} \longrightarrow \text{group solvability}}.$$

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